Associations Between Sleep Duration Across Development and Hippocampal Subfield Volumes at Midlife

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Background

- Past research has shown that multiple brain regions are susceptible to sleep deprivation and sleep disruption **[1]**
- Experimental studies in rodents have shown that poor sleep inhibits hippocampal neurogenesis and decreases hippocampal production of proteins linked to neuroplasticity **[2]**

Aims & Hypotheses

- **Aim:** To examine long-term effects of childhood and midlife sleep patterns on hippocampal subfield volumes at midlife
- **Hypothesis 1:** Poorer sleep during development will be associated with reduced hippocampal CA3 subfield volumes in adulthood
- **Hypothesis 2:** Poorer sleep during development will be associated with reduced hippocampal volumes in adulthood

Methods

Figure 1. *Timeline displaying CHDS and HBP study components **(above)**.*

Participants

- 91 middle-aged individuals from mother-offspring dyads who were enrolled in the Child Health and Development Studies (CHDS) **[3]** participated in the Healthy Brains Project (HBP) follow-up study **[4]**

Sleep Measures

- Childhood sleep duration was ascertained from an **interview battery** conducted with parents as part of the CHDS protocol when participants in the current study were 5 years old **[3]**
- Midlife sleep duration was examined using the **Pittsburgh Sleep Quality Index (PSQI)**, a self-report measure designed to evaluate subjective sleep quality within the past month **[5]**

MRI Data Preprocessing & Analysis

- Parcellation of the hippocampus and subfields (***Figure 2***) was performed using FreeSurfer, a software package that performs volumetric segmentation of macroscopically visible brain structures **[6-7]**

Results

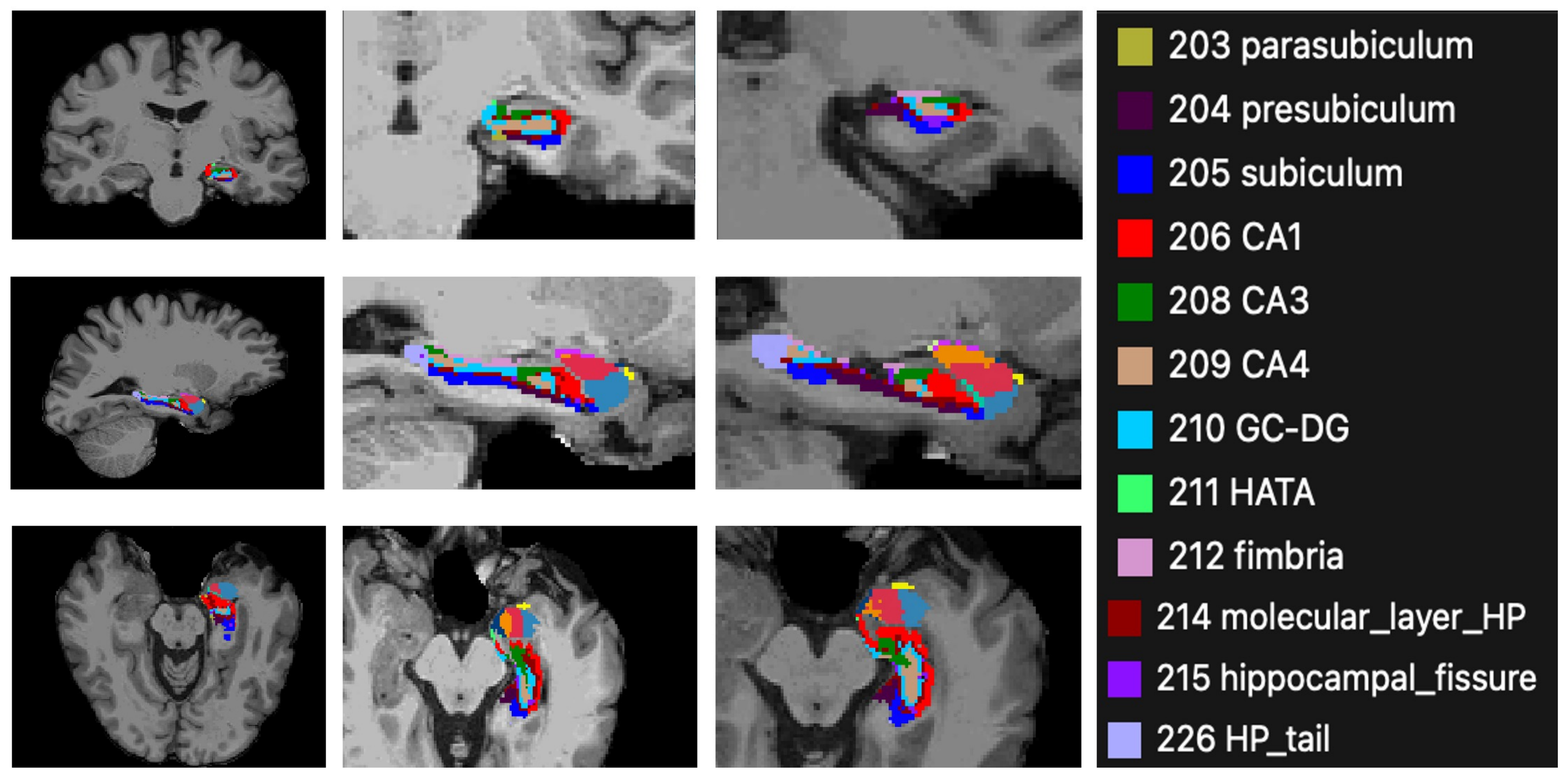

Figure 2. (Left) Visualizations of hippocampal subfield parcellations on representative coronal, axial, and sagittal planes. **(Right)** Hippocampal subfield color-coding key.

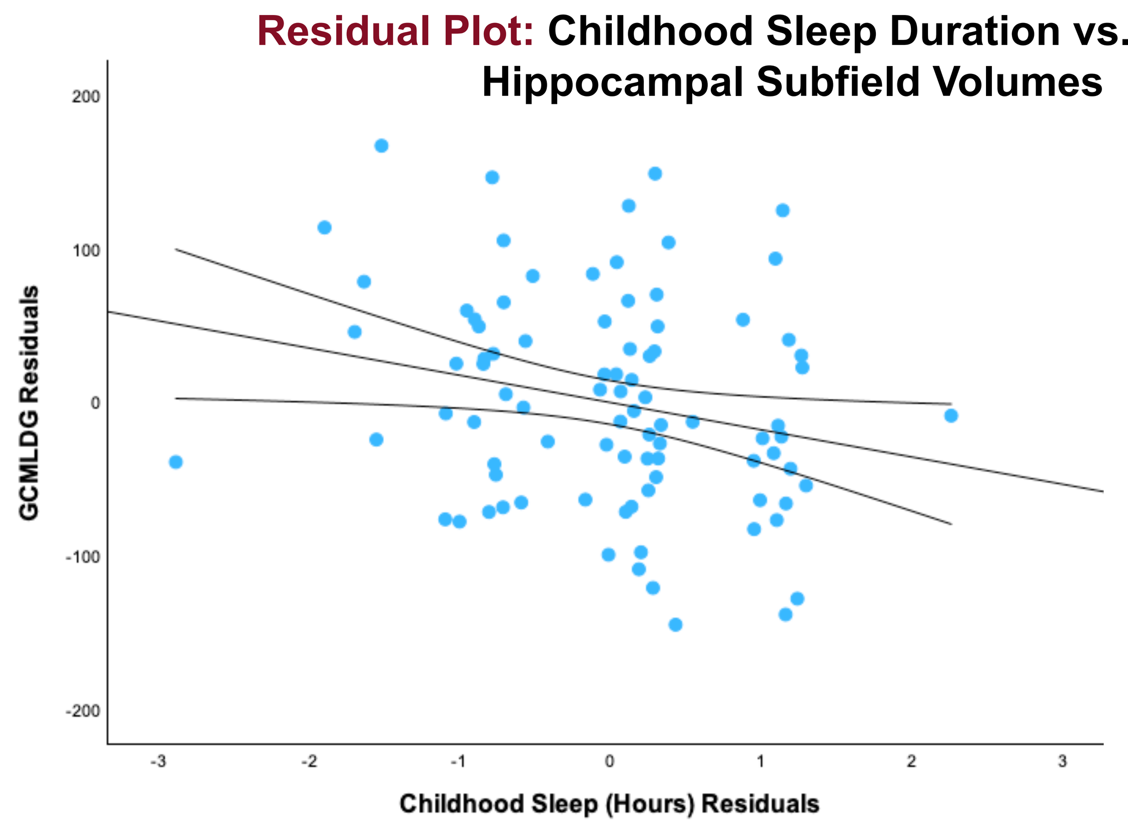

Figure 3. The significant partial correlation between **early life sleep duration** and **GCMLDG** volumes, controlling for ICV, visualized by plotting unstandardized residuals.

(r=-0.237, p=0.027)

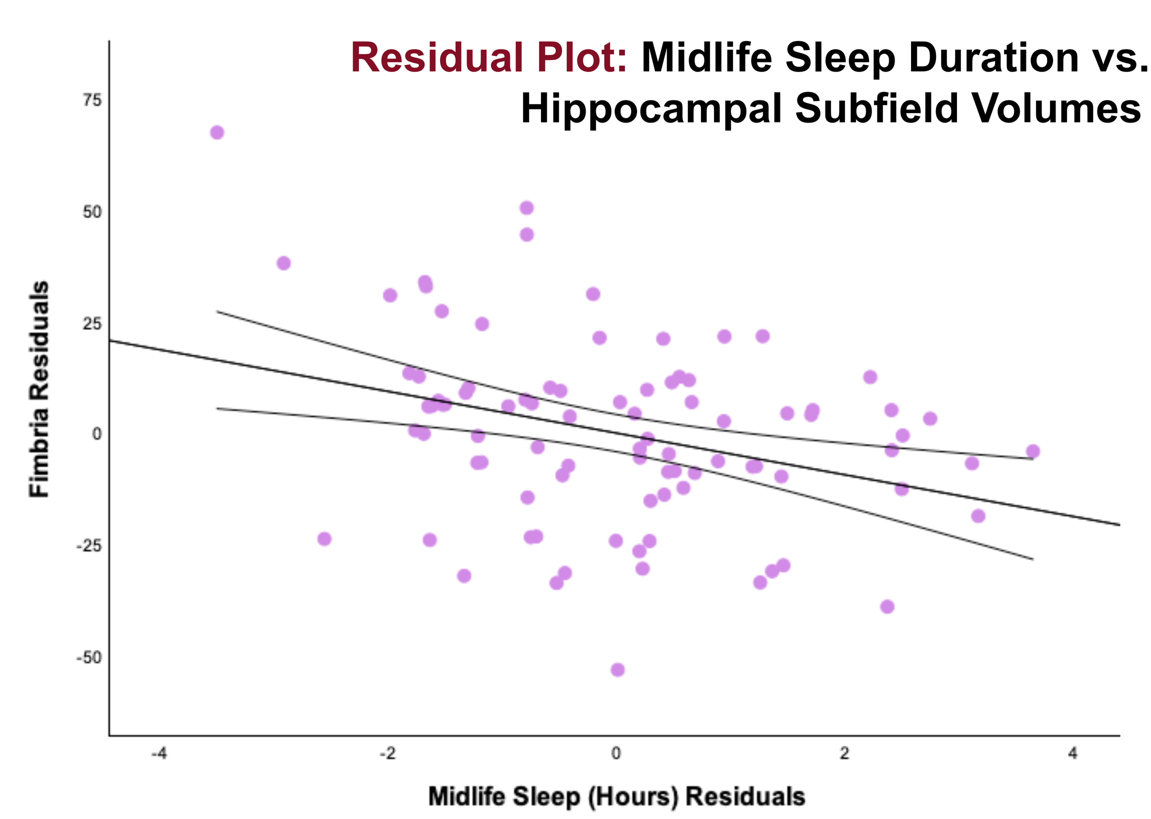

Figure 4. The significant partial correlation between **midlife sleep duration** and **fimbria** volumes, controlling for ICV, visualized by plotting unstandardized residuals.

(r=-0.288, p=0.007)

Participant Demographics

Table 1. Participant Demographics (n=91)	
Age (years) M (SD) [range]	59.45 (1.137) [57-63]
Sex % (n) Male	48.3 (44)
Ethnicity % (n) Hispanic	8.1 (7)
Race % (n)	
American Indian/Alaska Native	2.2 (2)
Asian	6.5 (6)
Black	12.1 (11)
White	68.1 (62)
More than one	8.7 (8)
Unknown	2.1 (2)

Discussion

- Findings suggest presence of differential influences of sleep duration on hippocampal subfield volumes across development
- Future studies of the Healthy Brains Project (HBP) cohort could benefit from examining cognitive functioning given typical age of onset for neurodegenerative disorders **[8]** and reports of increased sleep duration serving as an early marker of neurodegeneration **[9]**
- Results provide impetus to screen for, and treat, early sleep disturbances given potential for impacts on mental and physical wellbeing across the lifespan

References

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Acknowledgments

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